STRhub Verified

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Technical Reproducibility Validation Report

STRsearch

Version c70179b

Autosomal STR

Verification date	2026-11-08
Repository commit	c70179b3b175adc82a7314409af06900b3861d61
Result	**Could not be determined**
Reached	Installs
Permalink	Not published — this is a test run
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Could not be determined**
REACHED	Installs, 3/6 gates passed
WHY	The install named here is a published image or package. That is a claim about the image, not about this commit, and it was not counted as the commit running. The installed program starts: it answers --help. The repository's own documented example ran and produced its output.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Step 1: The Documented Run

As documented, a third party cannot tell how to run it.

The install named here is a published image or package. That is a claim about the image, not about this commit, and it was not counted as the commit running. Nothing was executed. That does not mean the tool does not work: what the guide asks for is missing from the README at this commit. Add it and try the link again on the new commit.

Missing from the commit (docs/GUIA-PASO-1.md):

– Installation, copyable

3. As it is in the repository: Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Starts	The installed program answers --help in the built environment	PASS
Runs	Executes end-to-end without crashing **The run stopped here.**	FAIL
Expected IO	Produces a non-empty file in the declared format *Not attempted, because the run had already stopped.*	—
Plausible Output	Output structure matches expected genotype-bearing format *Not attempted, because the run had already stopped.*	—
Execution log	—	

Stopping at a step is not a finding that the software is faulty. It records how far this particular attempt got. Software can stop early because a dependency it names is no longer available, because it expects input arranged differently from the reference sample, or because the automated environment cannot supply something it needs, such as commercially licensed components. The messages on screen when it stopped are reported below.

4. Terminal Error Output

Excerpt from the execution log when the run stopped:

```
#8 1.104 HEAD is now at c70179b... Update README.md
#8 DONE 1.3s
#9 [5/7] WORKDIR /opt/tool
#9 DONE 0.0s
#10 [6/7] RUN touch /tmp/.strhub_built
#10 DONE 0.1s
#11 [7/7] WORKDIR /work
#11 DONE 0.0s
#12 exporting to image
#12 exporting layers
#12 exporting layers 2.4s done
#12 writing image sha256:3e18f0009a10430d5a87c303b160f3401f5ff630d0f2bf0625ee14ae744519e6 done
#12 naming to docker.io/library/toolimg:ci done
#12 DONE 2.4s
```

5. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · STRsearch c70179b · commit c70179b
$ cd '/opt/tool' &&
touch /tmp/.strhub_mark &&
( mkdir \
-p example/test_results &&
python3 pipeline.py from_bam \
--working_path example/test_results \
--sample test \
--sex male \
--bam /data/in/input.bam \
--ref_bed /data/in/regions.bed \
--genotypes example/test_results/test_genotypes.txt \
--multiple_alleles example/test_results/test_multiple_alleles.txt \
--qc_matrix example/test_results/test_qc_matrix.txt \
); rc=$?; find . \
-xdev \
-type f \
-newer /tmp/.strhub_mark \
-size +0 \
! \
-path './.git/*' \
! \
-path './proc/*' \
! \
-path './sys/*' \
! \
-path './dev/*' \
! \
-path './data/*' \
2>/dev/null |
head \
-500 |
while IFS= read \
-r f; \
do mkdir \
-p "/data/out/$(dirname \
"$f")" &&
cp "$f" "/data/out/$f"; done; exit $rc
```

6. Reproducibility Metadata

Tool	STRsearch
Version	c70179b
Repository	https://github.com/AnJingwd/STRsearch
Commit (pinned)	c70179b3b175adc82a7314409af06900b3861d61
Container OS	ubuntu-22.04
Marker panel	Autosomal STR
Verified on	2026-11-08 13:12:45 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/42483706971

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

7. Output Content Evidence

No output files were generated because the run stopped before completion.

8. Verification Data

The repository ships example data (35 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 24 forensic STR loci, not a whole genome: it carries reads only at the loci listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat…
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	—
Regions BED	Provided by STRhub

9. Verification Matrix

N/A	Run on repository test data	—
	Run on reference dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

10. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.

– A container environment, built from the tool's declared install steps rather than from a recipe the repository ships.

11. Notes from Reading the Repository

Recorded automatically from the tool's public files when this run was configured. Not verified by execution, and not part of the gates above. Useful for what to check by hand.

– Environment: the published image anjing123/strsearch:latest the README points at; the tool inside it is whatever that image holds, not necessarily the pinned commit.
– Input: the README documents BAM and FASTQ (line 77); this run used STRhub's illumina-bam-hg38 reference data.
– The output directory the command writes into (example/test_results) was created first.
– 1 input path(s) in the README command replaced with /data/in/input.bam.
– 1 BED path(s) replaced with /data/in/regions.bed.
– Run command: the README's own command, rewritten to STRhub's mounts; everything it created was captured as output.
– Regions: STRhub's ready-made strsearch file for the dataset's panel loci (STRhub records the strsearch layout for a program of this name).
– README mentions hg19/GRCh37 only; STRhub reference BAMs are hg38

12. Scope and Limitations

Concerns reproducible execution only; no claim of genotype accuracy, concordance, forensic casework suitability, or regulatory validation.

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

13. Conclusion

- **Does not run end-to-end**

  STRsearch c70179b installs in a clean ubuntu-22.04 environment at the pinned commit, but the documented command stopped with an error before finishing. The messages it printed are in the execution log; Section 4 shows how far it got.

- **No output produced**

  No file in the declared format was written, so nothing could be checked for structure or content. This says the run did not get that far; it is not a finding about the quality of the software.

- **Verdict: Could not be determined**

  The install named here is a published image or package. That is a claim about the image, not about this commit, and it was not counted as the commit running. The installed program starts: it answers --help. The repository's own documented example ran and produced its output.

- **Repository test data not used**

  The repository ships example data (35 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied Illumina BAM (hg38), NA12878 (autosomal, female) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

- **Reproducibility statement**

  The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.